

- Q.29 Describe any three types of filters in brief.
 Q.30 Differentiate micro filtration and ultra filtration.
 Q.31 What is the impact of size reduction on food.
 Q.32 What are the basic principles of centrifugation, and how is it applied in the food industry?
 Q.33 Describe the factors affecting the size reduction of food.
 Q.34 Explain the mechanical separation process.
 Q.35 Differences between batch and continuous mixing, and its benefits and drawbacks.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the application of the extraction process in food industries.
 Q.37 Explain the types of membrane separation process.
 Q.38 Discuss the role of filtration in food processing and mention different types of filtration methods used.

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Roll No.

3rd Sem / Food Technology Sub : Unit Operation in Food Processing / Pr. Of food Engg.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is NOT a mixing method?
 a) Filtration b) Stirring
 c) Shaking d) Agitation
- Q.2 What type of operation is involved in separating liquids from solids?
 a) Mixing b) Centrifugation
 c) Homogenization d) Extrusion
- Q.3 Gravity separator makes use of the difference in
 a) Moisture content b) Electrical properties
 c) Magnetic properties d) Densities
- Q.4 Which of the following equipment is used for solid-liquid extraction?
 a) Tray dryer b) Screw press
 c) Heat exchanger d) Grinder
- Q.5 Which of the following is an example of a unit operation used to improve the shelf life of a food product?
 a) Dehydration b) Mixing
 c) Grinding d) Blending

- Q.6 Which process involves passing a liquid through a porous material to remove particle?
 a) Distillation b) Filtration
 c) Homogenization d) Crystallization
- Q.7 Which of the following is an example of a membrane separation process?
 a) Ultra filtration b) Centrifugation
 c) Spray drying d) Sedimentation
- Q.8 Which of the following is NOT a type of size reduction equipment?
 a) Crusher b) Centrifuge
 c) Grinder d) Hammer mill
- Q.9 The process of mixing two immiscible liquids to form a stable mixture is called;
 a) Fermentation b) Emulsification
 c) Dehydration d) Crystallization
- Q.10 Which of the following is a form of mechanical separation?
 a) Ultra filtration b) Centrifugation
 c) Spray drying d) Homogenization

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 _____ used to reduce particle size in liquid food.
- Q.2 The purpose of grading and sorting is the same (True/False)

- Q.13 _____ mill used for milling fibrous food products.
- Q.14 Distillation separate liquid based on the difference in specific gravity (True / False)
- Q.15 Define sublimation.
- Q.16 The main aim of emulsification is to kill pathogenic microorganisms (True / False)
- Q.17 _____ law is used to determine energy requirement for coarse grinding.
- Q.18 What is filter media.
- Q.19 Mention any two types of sieves.
- Q.20 Two examples of filter aids_____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the working of the hammer mill with a neat diagram.
- Q.22 Explain any four types of screens used in the food industry.
- Q.23 Explain the physical separation process and its application.
- Q.24 Write about steam distillation with the help of a neat diagram.
- Q.25 Explain the working principle of filtration.
- Q.26 Discuss any two types of mixers in brief.
- Q.27 Write a brief note on dry cleaning methods used in the food industry.
- Q.28 Explain the different stages of ice crystal formation.